

ZVERINOGOLOVSKOYE, Russian Federation

WMO: 287560

Lat: 54.467N

Lon: 64.867E

Elev: 384

StdP: 14.49

Time Zone: 5.00 (E05)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.4	-21.0	-31.8	0.9	-26.2	-26.3	1.3	-20.8	22.4	1.0	19.1	2.2	3.3	180	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	91.0	69.1	87.5	67.8	84.0	66.3	72.4	85.0	70.2	83.1	68.2	80.0	6.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.2	104.9	79.6	65.9	96.7	75.6	63.8	90.0	74.1	36.3	84.8	34.4	83.6	32.7	80.0	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.3	16.5	14.2	-33.1	96.3	6.6	3.0	-37.9	98.5	-41.7	100.2	-45.4	101.9	-50.2	104.1		
(5)			-33.3	74.8	6.6	3.7	-38.0	77.5	-41.8	79.6	-45.5	81.7	-50.3	84.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.2	-0.1	4.7	19.3	41.3	56.6	65.6	68.6	64.5	54.2	39.2	23.1	6.9		
	DBStd	26.42	11.46	12.87	10.42	10.58	9.04	8.58	7.15	7.95	9.16	9.19	11.45	13.87		
	HDD50	6670	1552	1269	951	301	39	6	0	2	59	350	806	1335		
	HDD65	10563	2017	1689	1416	713	285	95	43	110	340	799	1256	1800		
	CDD50	1988	0	0	0	39	245	475	577	451	185	16	0	0		
	CDD65	409	0	0	0	1	26	114	156	95	17	0	0	0		
	CDH74	4733	0	0	0	21	407	1415	1718	980	192	0	0	0		
	CDH80	1744	0	0	0	3	113	539	687	346	56	0	0	0		
	WSAvg	6.1	6.1	6.5	7.2	7.1	6.7	5.9	5.6	4.9	5.1	5.7	6.3	6.4		
	PrecAvg	14.3	0.7	0.6	0.7	0.8	1.3	1.7	2.2	1.9	1.2	1.1	1.1	0.9		
	PrecMax	22.8	1.5	1.7	1.7	1.9	4.5	5.0	5.2	4.7	2.2	3.0	2.0	2.2		
(15) Precipitation	PrecMin	7.4	0.1	0.1	0.0	0.1	0.1	0.2	0.2	0.7	0.0	0.1	0.2	0.1		
	PrecStd	4.2	0.4	0.4	0.4	0.5	1.1	1.3	1.3	1.0	0.7	0.6	0.5	0.5		
	0.4% DB	30.0	35.6	48.2	76.8	88.6	93.7	96.7	92.8	86.7	68.5	47.9	35.5			
	0.4% MCWB	28.4	32.5	42.1	57.1	65.4	68.7	69.7	72.5	63.3	53.6	43.3	32.5			
	2% DB	26.0	32.3	38.6	69.5	83.1	90.0	91.3	88.1	79.4	62.6	42.7	31.9			
(21) Mean Coincident Wet Bulb Temperatures	2% MCWB	24.8	30.5	35.4	53.5	62.0	67.7	69.5	69.0	60.3	51.0	40.1	30.6			
	5% DB	22.1	28.3	35.8	64.1	78.6	86.5	87.2	83.7	74.3	58.0	39.5	28.8			
	5% MCWB	20.9	26.8	33.3	50.8	61.0	67.2	68.9	67.2	59.1	48.6	37.0	27.3			
	10% DB	17.6	23.3	33.8	58.7	74.3	82.4	83.4	79.5	69.3	53.0	36.3	25.5			
	10% MCWB	16.5	21.8	31.7	48.2	58.7	65.3	67.4	65.3	57.0	46.1	34.2	24.1			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	28.9	33.0	42.9	58.3	68.2	74.7	74.9	75.6	67.4	55.1	44.1	32.8			
	0.4% MCDB	29.7	34.8	48.2	73.5	81.8	87.9	86.1	88.6	82.2	65.1	46.7	34.6			
	2% WB	25.0	30.5	35.5	54.5	64.4	70.7	72.2	72.1	62.5	52.2	40.3	30.3			
	2% MCDB	26.1	32.4	37.7	67.0	77.9	85.2	85.3	83.7	74.1	61.1	42.5	31.9			
	5% WB	21.0	26.9	33.6	51.5	62.1	68.2	70.5	68.6	60.2	49.2	37.3	27.3			
(31) Mean Daily Temperature Range	5% MCDB	22.1	28.2	35.5	62.7	76.6	82.0	83.3	79.7	72.1	56.9	39.4	28.8			
	10% WB	16.6	21.9	31.8	48.7	59.6	66.2	68.9	66.2	57.6	46.5	34.4	24.1			
	10% MCDB	17.6	23.2	33.6	58.3	72.6	80.2	80.5	76.4	68.1	52.1	36.1	25.4			
	5% DB	13.7	15.7	15.6	18.5	22.8	22.0	20.7	20.9	19.4	15.9	11.6	13.5			
	5% MCDBR	15.9	16.1	13.3	27.4	29.8	27.9	25.9	26.7	25.9	23.8	11.7	13.8			
(36) Clear-Sky Solar Irradiance	5% WB	15.5	14.7	10.9	14.6	14.1	13.6	11.4	13.3	13.1	14.4	9.5	12.9			
	5% MCDBR	15.9	15.8	12.8	26.0	26.8	25.3	23.2	24.0	23.6	22.3	10.9	13.5			
	5% WB	15.5	14.7	10.9	14.6	14.1	13.6	11.4	13.3	13.1	14.4	9.5	12.9			
	taub	0.289	0.322	0.375	0.411	0.411	0.408	0.426	0.440	0.385	0.359	0.320	0.290			
	taud	2.058	2.015	1.955	2.124	2.217	2.300	2.247	2.179	2.287	2.243	2.157	2.041			
(42) All-Sky Solar Radiation	Ebn at Noon	196	231	245	256	264	266	258	245	245	220	184	165			
	Edh at Noon	22	33	44	43	42	39	40	41	32	26	20	18			
	RadAvg	322	669	1161	1465	1790	1932	1807	1464	1026	528	268	211			
(45)	RadStd	22	24	76	157	158	158	153	142	74	78	29	24			

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page